



Use the papers provided by the department. Write clearly with short and concise motivations. Illustrate with a figure when necessary. The cover shall be filled in completely. Write initials on each page you hand in.

1. Consider the series

$$s(x) = \sum_{n=1}^{\infty} \frac{x^n}{n}.$$

Find the radius of convergence R and compute the value of $s(x)$ when $|x| < R$. Finally, determine what can be said about convergence when $x = -R$ and $x = R$, respectively.

2. Solve the following problem for the heat equation

$$\begin{cases} u_t = u_{xx}, & 0 < x < \pi, t > 0, \\ u_x(0, t) = u_x(\pi/2, t) = 0, & t > 0, \\ u(x, 0) = x(\pi - x), & 0 < x < \pi. \end{cases}$$

3. Consider the 2π -periodic function $f(x) = \max\{\cos x, 0\}$, $x \in \mathbb{R}$. Sketch the graph in the interval $-\pi \leq x \leq \pi$ and find the Fourier series. Use the result to compute the sum

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}.$$

4. Consider the function

$$f(x) = \begin{cases} \sin x & |x| \leq \pi \\ 0 & |x| > \pi \end{cases}.$$

Compute the Fourier transform $\hat{f}(\xi)$ and determine the integral

$$\int_{-\infty}^{+\infty} \frac{\sin^2(\pi x)}{(x^2 - 1)^2} dx.$$

5. Consider the functions $f_n(x) = 2nx(1 - x^2)^{n-1}$.

(a) Prove that the sequence $(f_n)_{n=1}^{\infty}$ is uniformly convergent on each interval $[\delta, 1]$ where $0 < \delta < 1$.

(b) Compute the limit

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) g(1 - x) dx$$

where g is a continuous function on $[0, 1]$.

6. (a) Calculate the Fourier transform of the function h where $h(x) = e^{-x}$ for $x \geq 0$ and $h(x) = 0$ for $x < 0$.

(b) Find a solution $f(x)$ to the integral equation

$$f(x) = e^{-|x|} + \frac{1}{2} e^x \int_x^{\infty} e^{-y} f(y) dy, \quad (x \in \mathbb{R}).$$